

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	19 April 2016	First Release	Chin Chee Yan

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

Content

Removal Procedure.....	3
Replacement Procedure.....	4
Belt Alignment Verification.....	7
PIP Sensor Flag Verification and Adjustment.....	12

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

Removal Procedure

1. Unload board from machine if any.
2. Go to X ray source tab and toggle “Turn off X-Ray” .Power off at Operator Console Panel

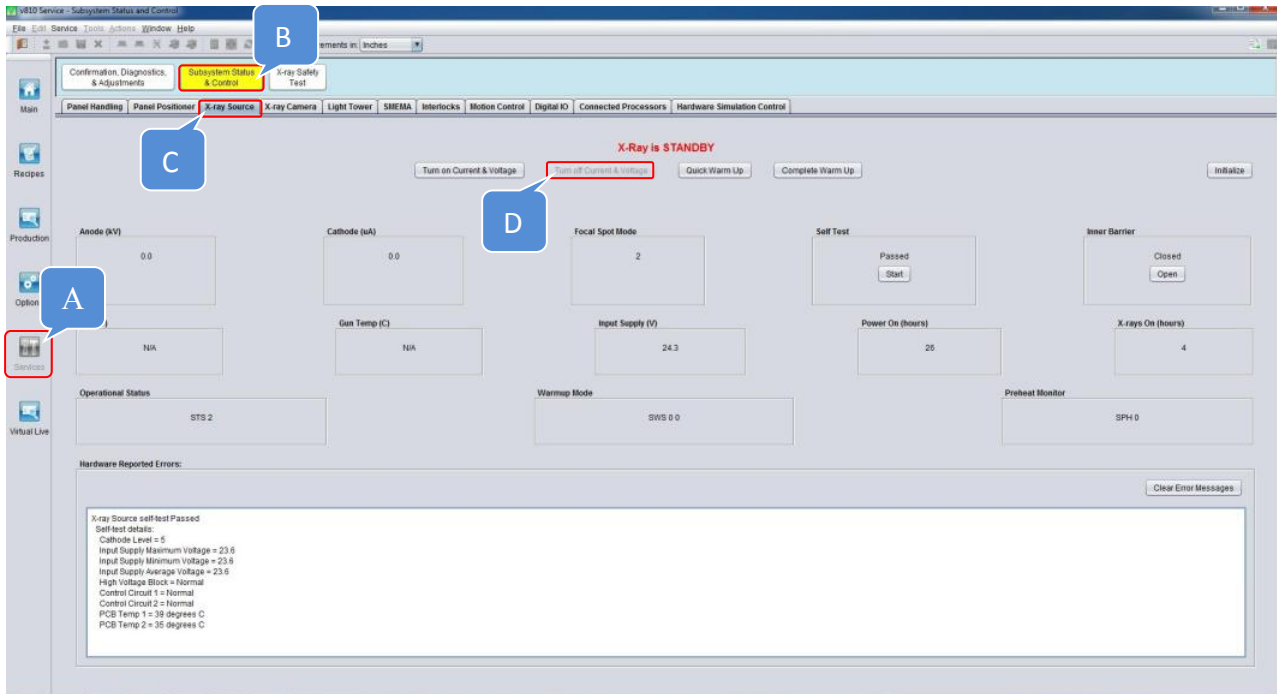


Figure 1 X ray Source

3. At Panel Handling tab, manually set the Rail Width to 8 inch
Services → Subsystem Status & Control → Panel Handling → Display unit to Inch → Initialize → 8 inch width → Open Left and right Outer Barrier → Open Front Access Door

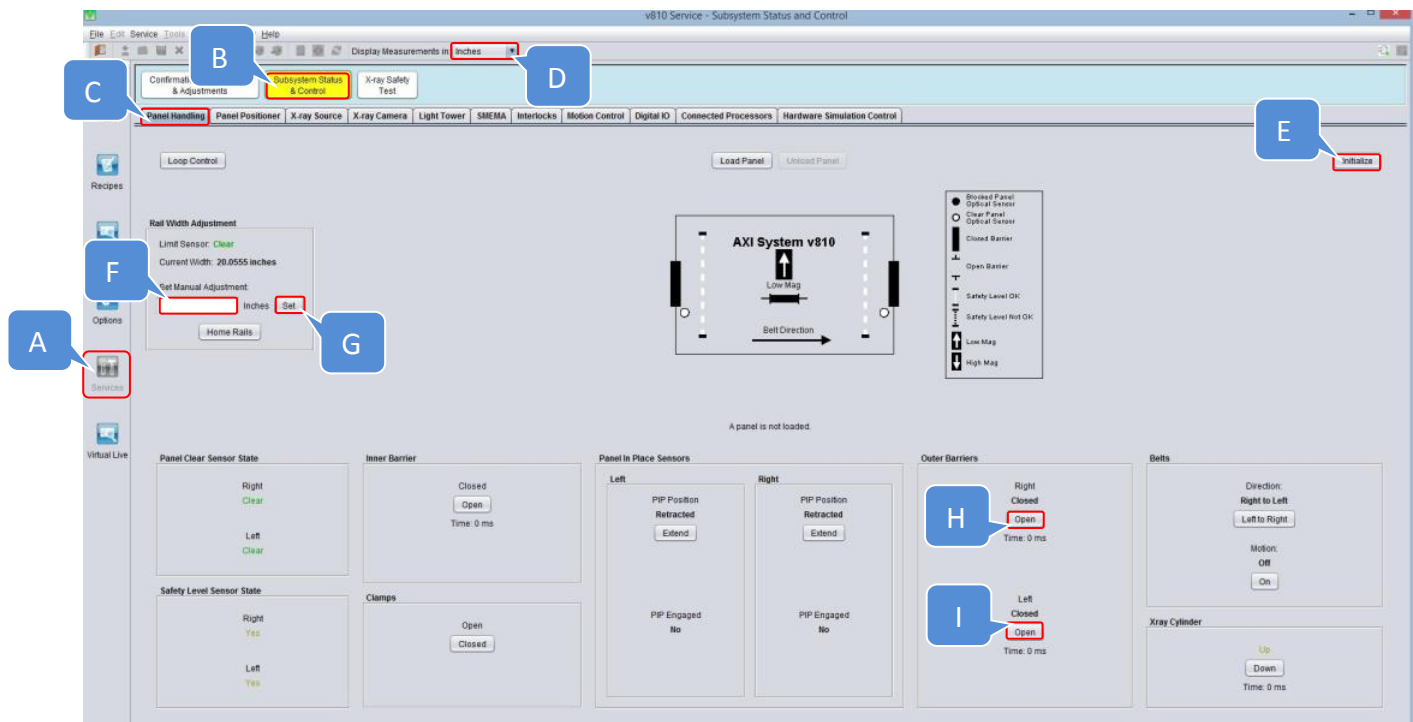
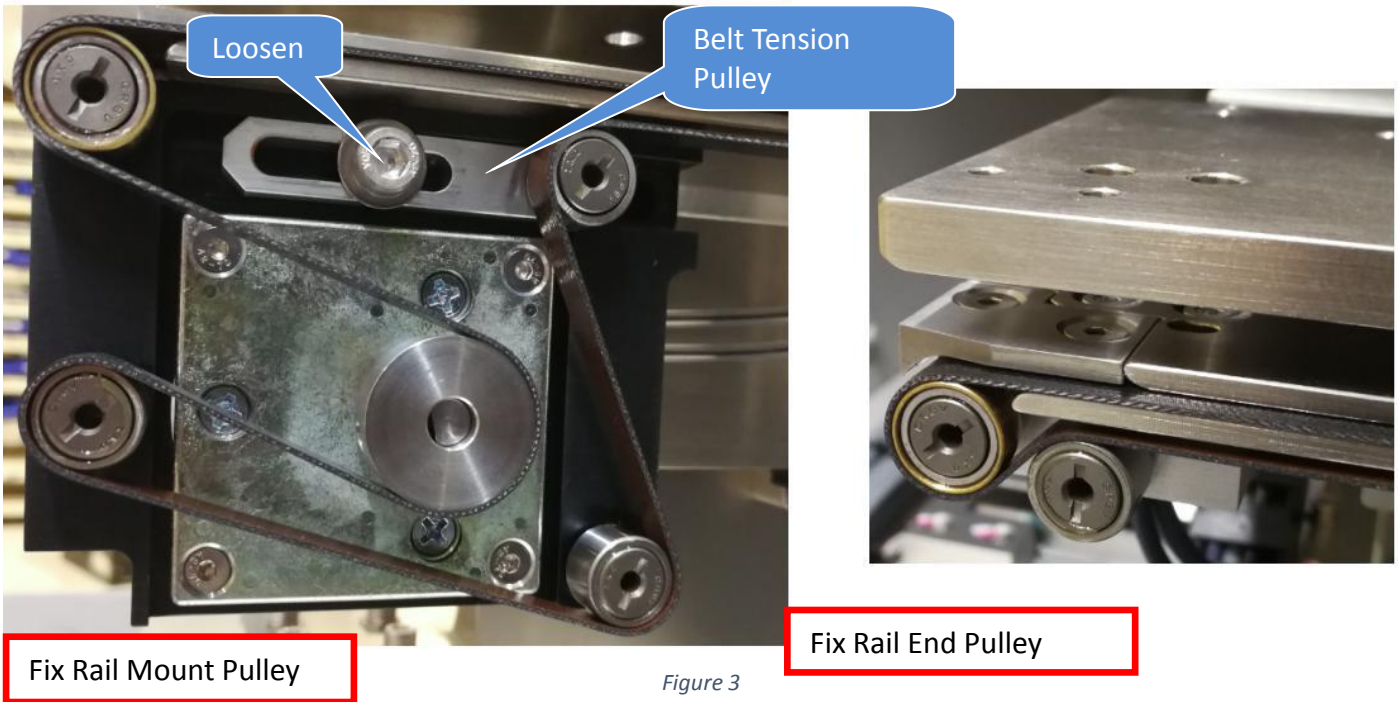


Figure 2 Rail Width Set

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

4. Manually move XY stage. Loosen screw at Belt Tension Pulley at Fix Rail. Remove Conveyor Belt from Fix Rail Mount Pulley Assembly & Fix Rail End Pulley Assembly



Replacement Procedure

5. Prepare new V810 Belt Replacement Kit (P/N, 4XASP-A025) as shown.

Remarks : 1 Kit Contain 2 pcs of Conveyor Belt



Figure 4 V810 Std/S2EX Conveyor Belt

ViTrox 	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

6. Route Conveyor as below Routing Diagram. Hand Tight Tension Pulley

Remarks :Belt Surface with Black Colour require to face upwards



Figure 5 V810 S2EX Conveyor Belt Routing

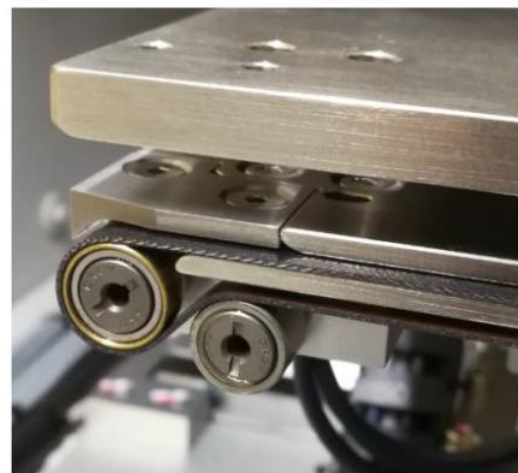
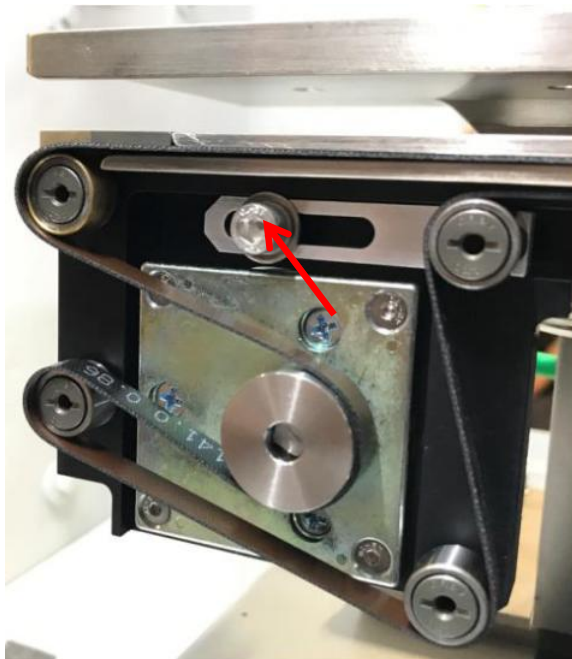


Figure 6

7. Visual Check Conveyor Belt where slot firmly along the track as below Figure 7

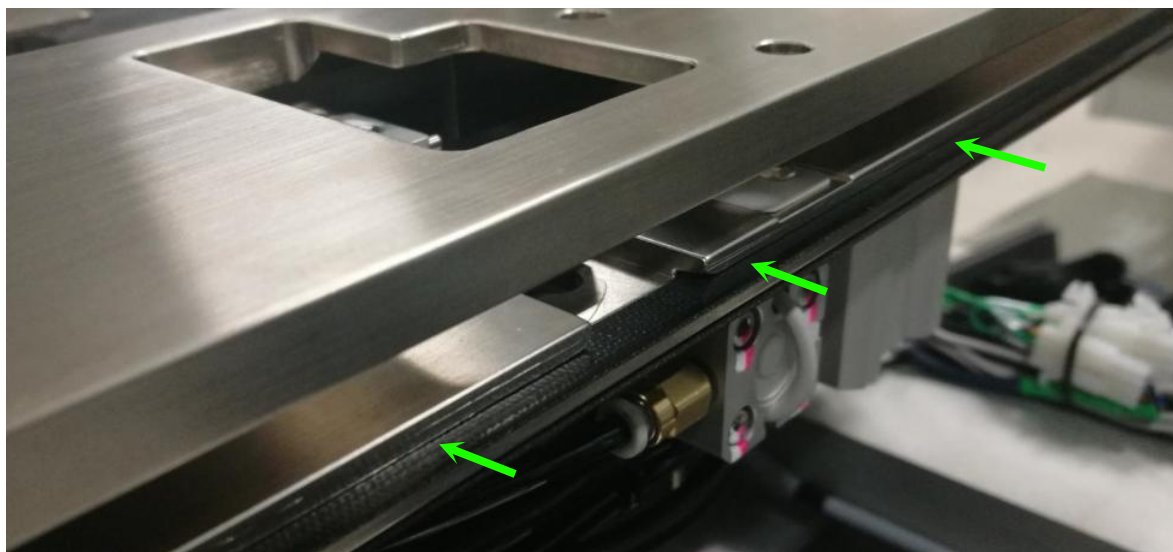


Figure 7

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

8. Prepare Belt Tension Kit (P/N, 4XASP-A114) as below Figure 8



Adjustable knob

Figure 8 Belt Tension Kit

9. Place Belt Tension Kit as below step

- a) Identify the Tiny Opening at Fix Rail Mount Pulley

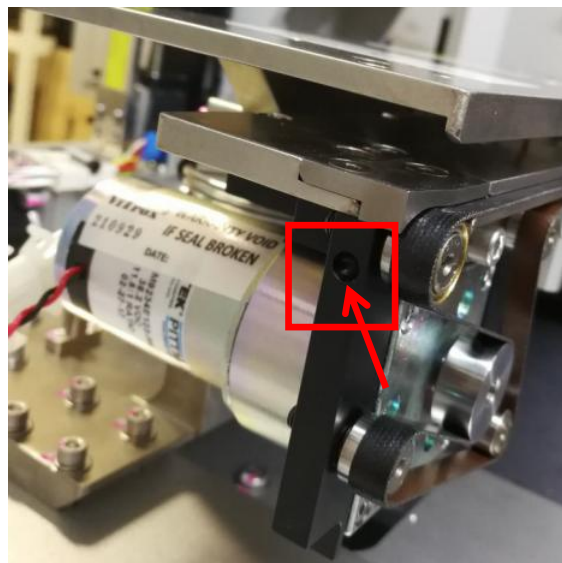
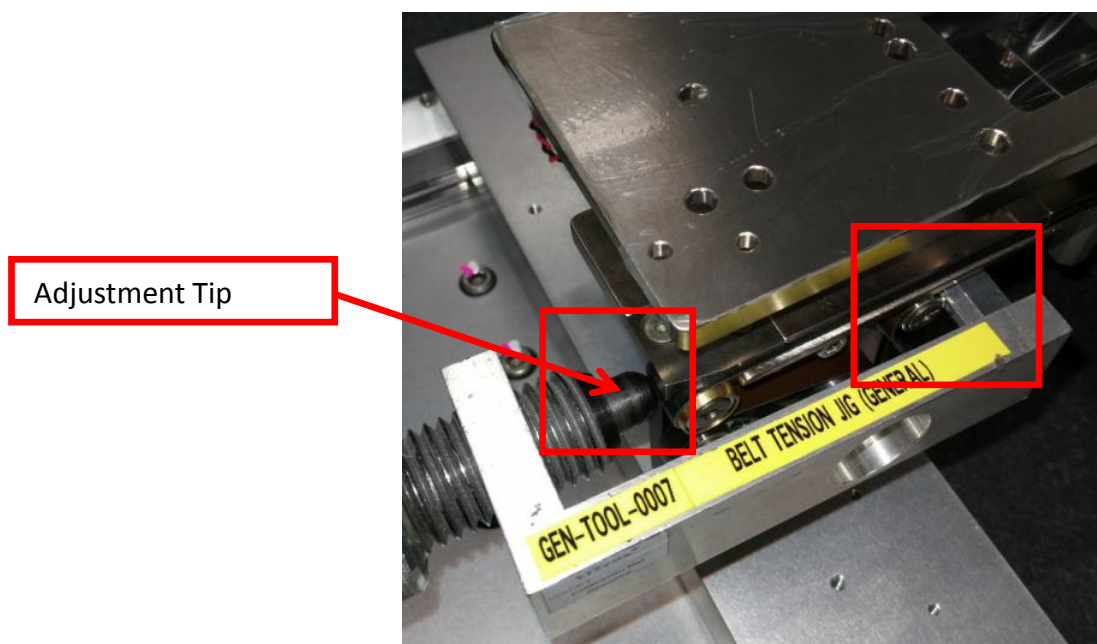


Figure 9 Tiny Opening

- b) Gently rest Belt Tension Adjustment Tip on Tiny Opening as below Figure 10 ,while other end will require to hook with the Tension Pulley.



Adjustment Tip

Figure 10 Belt Tension Kit Placement

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

10. Loosen the Belt Tension Pulley Block screw. Gently adjust Adjustment Knob until Tension Block is within 3/4 adjustment scale. Then, tighten the Belt Tension Pulley Block Screw

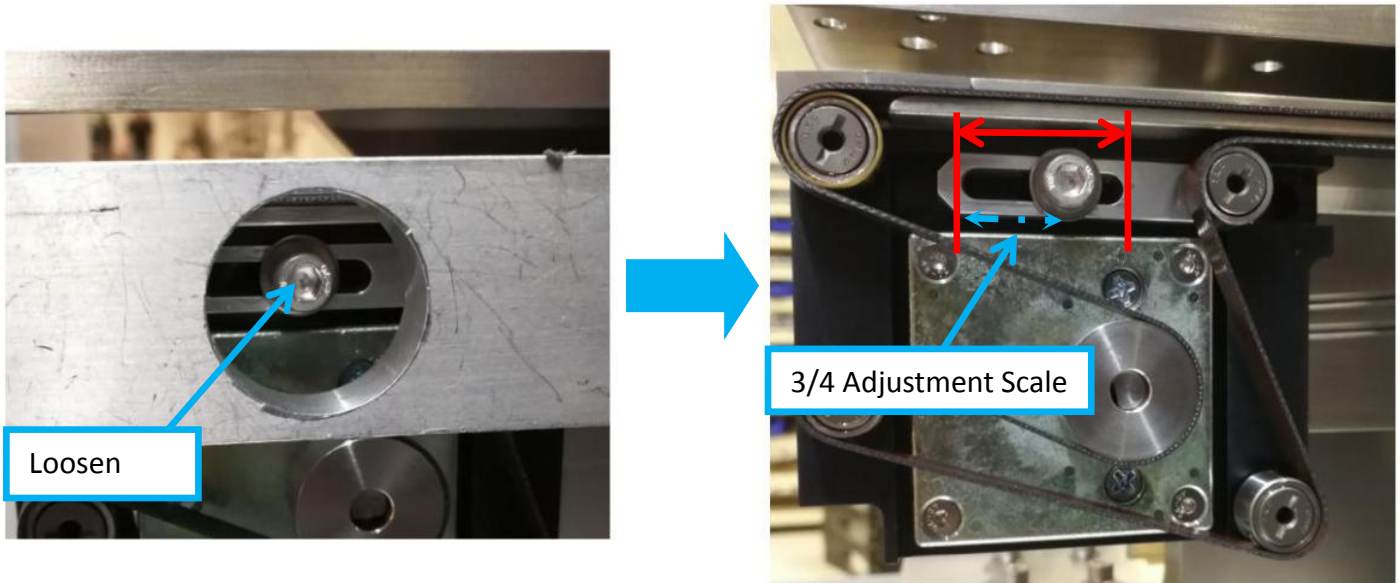


Figure 11 Belt Tension Pulleys

11. Release and remove Belt Tension Kit & tools from machine. Manually move the X axis towards Left Outer Barrier Loading direction

Belt Alignment Verification

12. Turn on Conveyor Motor at Digital IO Tab (Stage belt Motor ON).

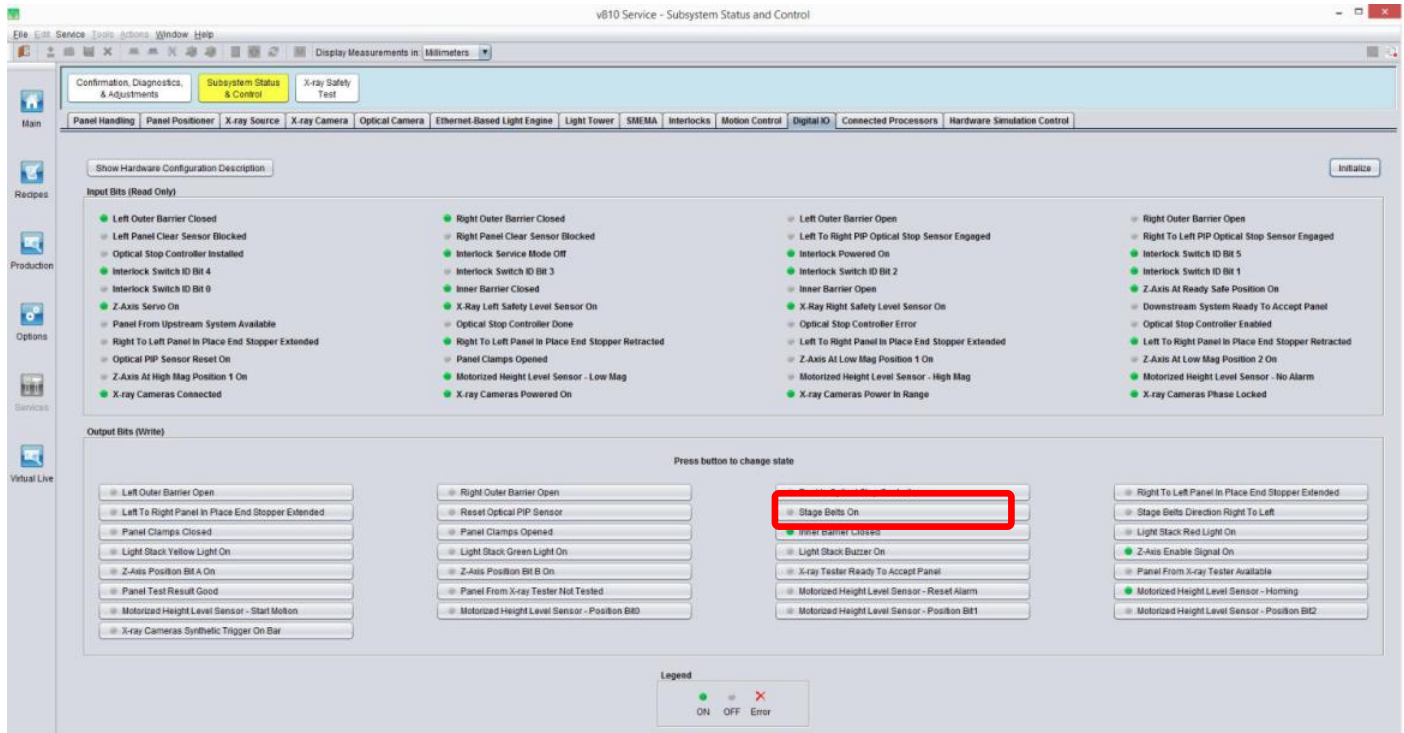


Figure 12 Belt Motor Toggle

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

13. Visual check Conveyor Belt Running condition

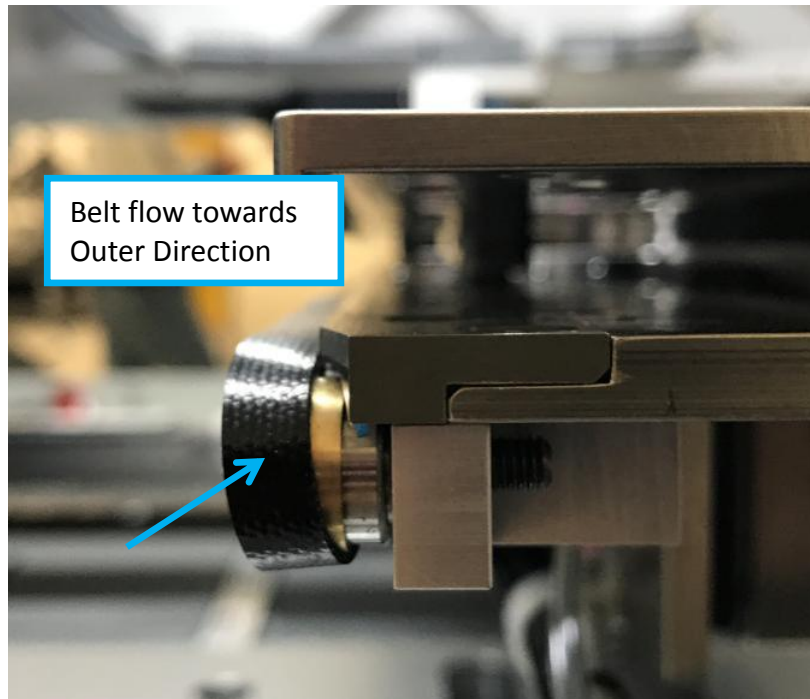


Figure 13 Belt Flow (NG)

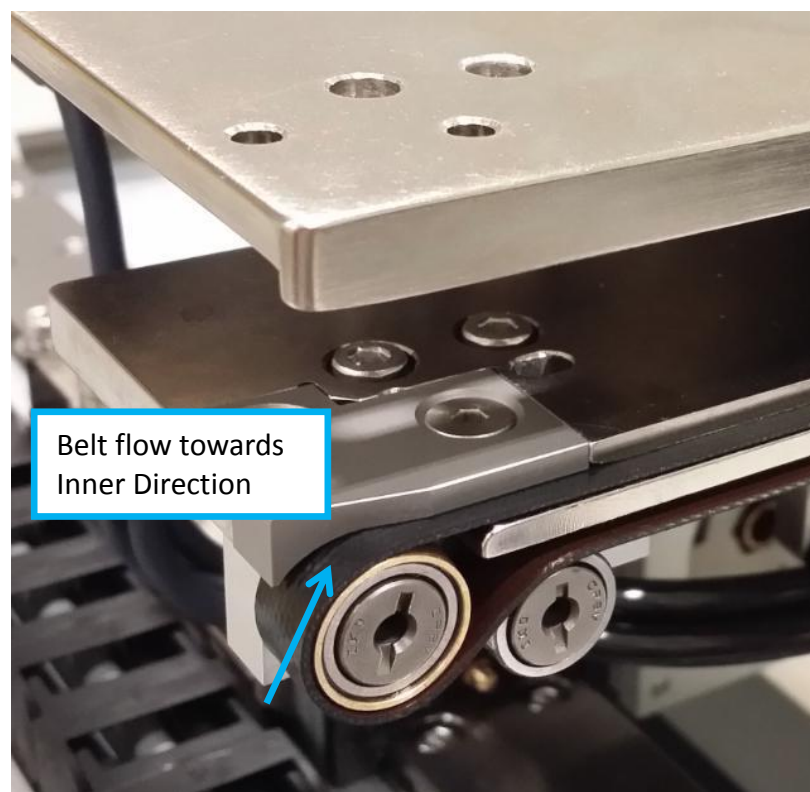


Figure 14 Belt Flow (Good)

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

14. Turn off the Belt Motor Motion at Digital IO Tab
15. Perform pulley alignment as below steps
 - a) Loosen 4 x Screws at Mount Pulley assembly as below Figure 15

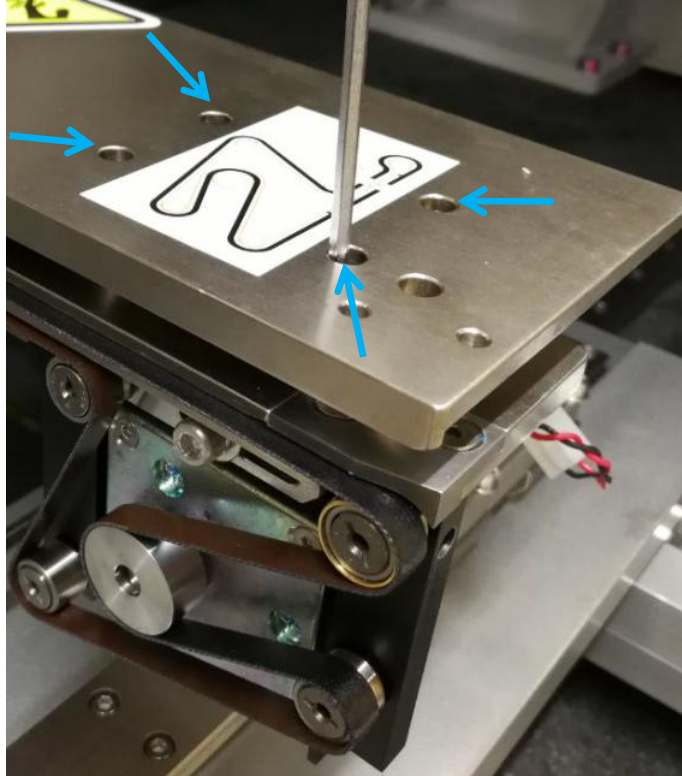


Figure 15 Fix Rail Mount Pulley Assembly

- b) Place L-ruler & Align Mount Pulley assembly as below Figure 16. Align Mount Pulley perpendicular with Fix Rail. Next, Tighten 4 x Screws

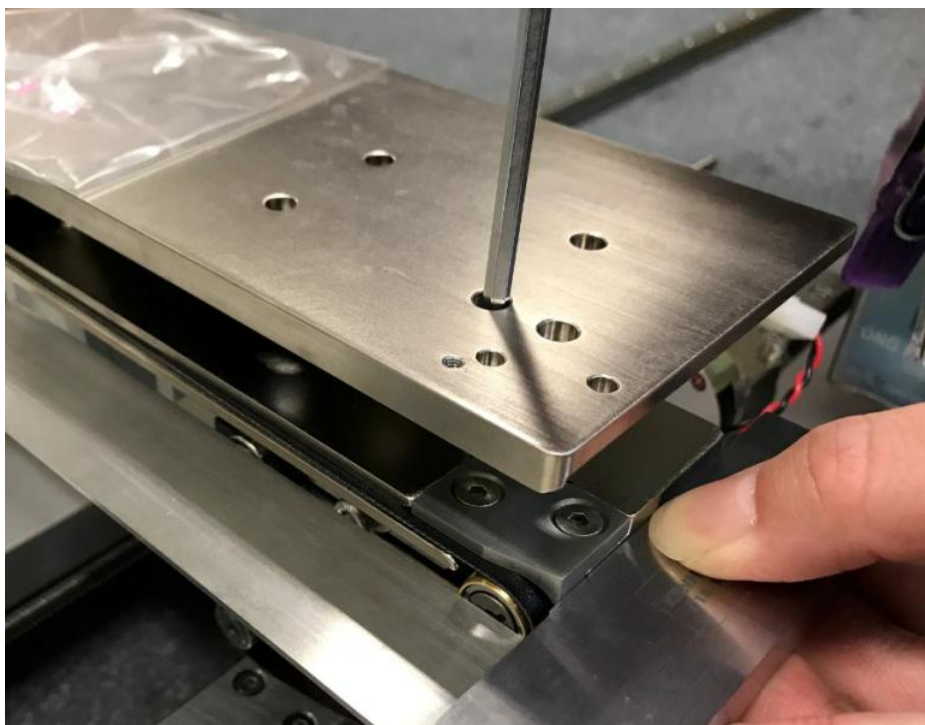


Figure 16 Fix Rail Mount Pulley Alignment

ViTrox 	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

- c) Toggle “Stage belt Motor ON “at Digital IO tab.
- d) Slightly loosen End Pulley assembly as below Figure 17

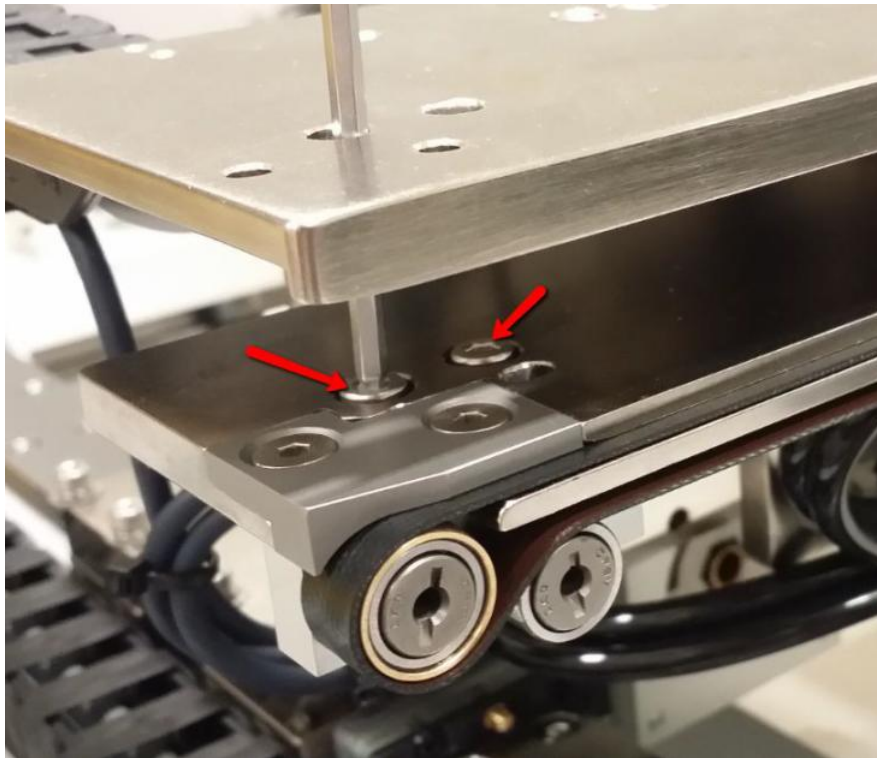


Figure 17 End Pulley Assembly

- e) Insert Allen Key to End Pulley as below Figure 18

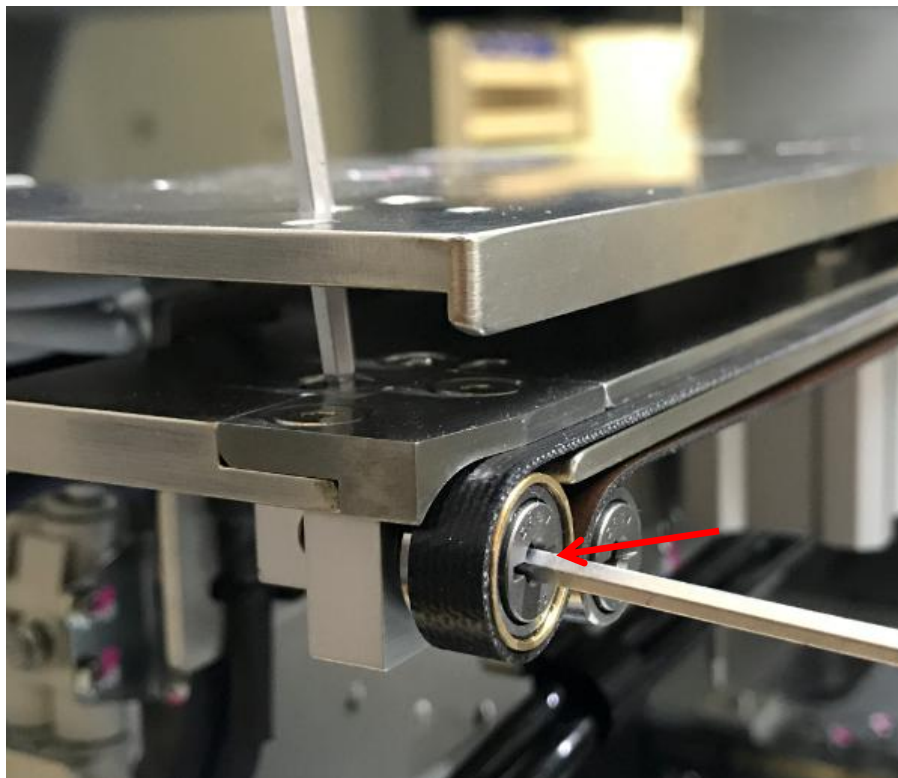


Figure 18 End Pulley Assembly

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

- f) Align the End Pulley until Belt Flow towards Inner Direction as below Figure 19. Tighten 2 x Screws after desire alignment achieve

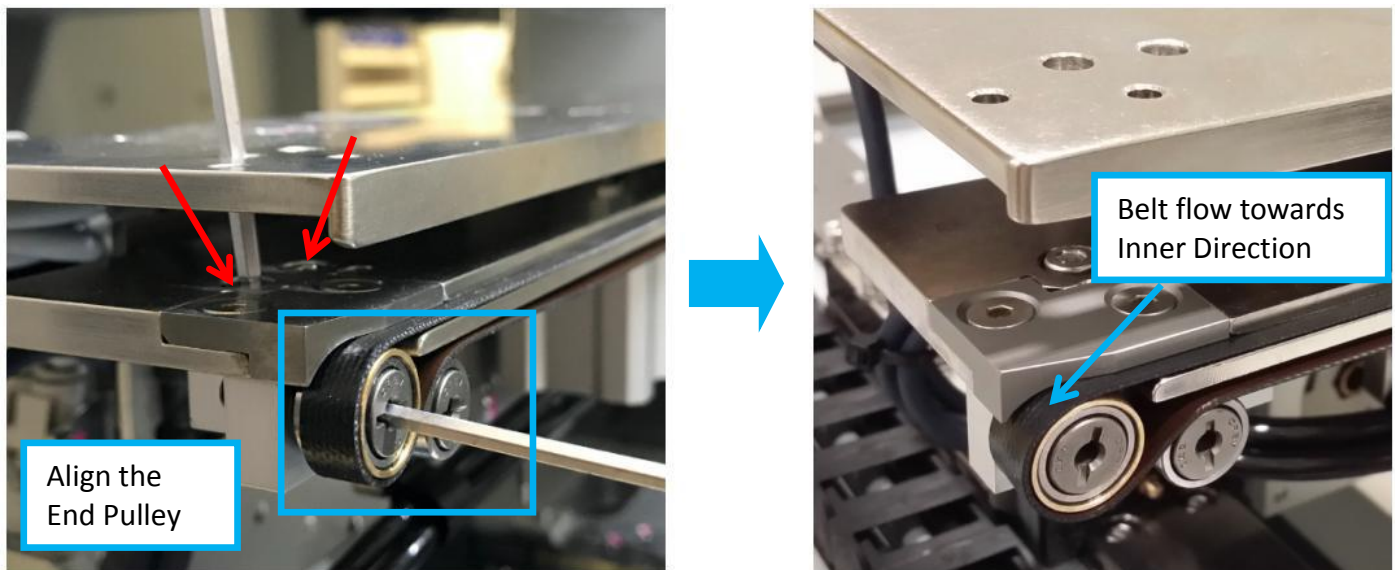


Figure 19 End Pulley Alignment

16. Toggle “Stage Belts Direction Right to Left “ while “Stage Belts On” remain ON.

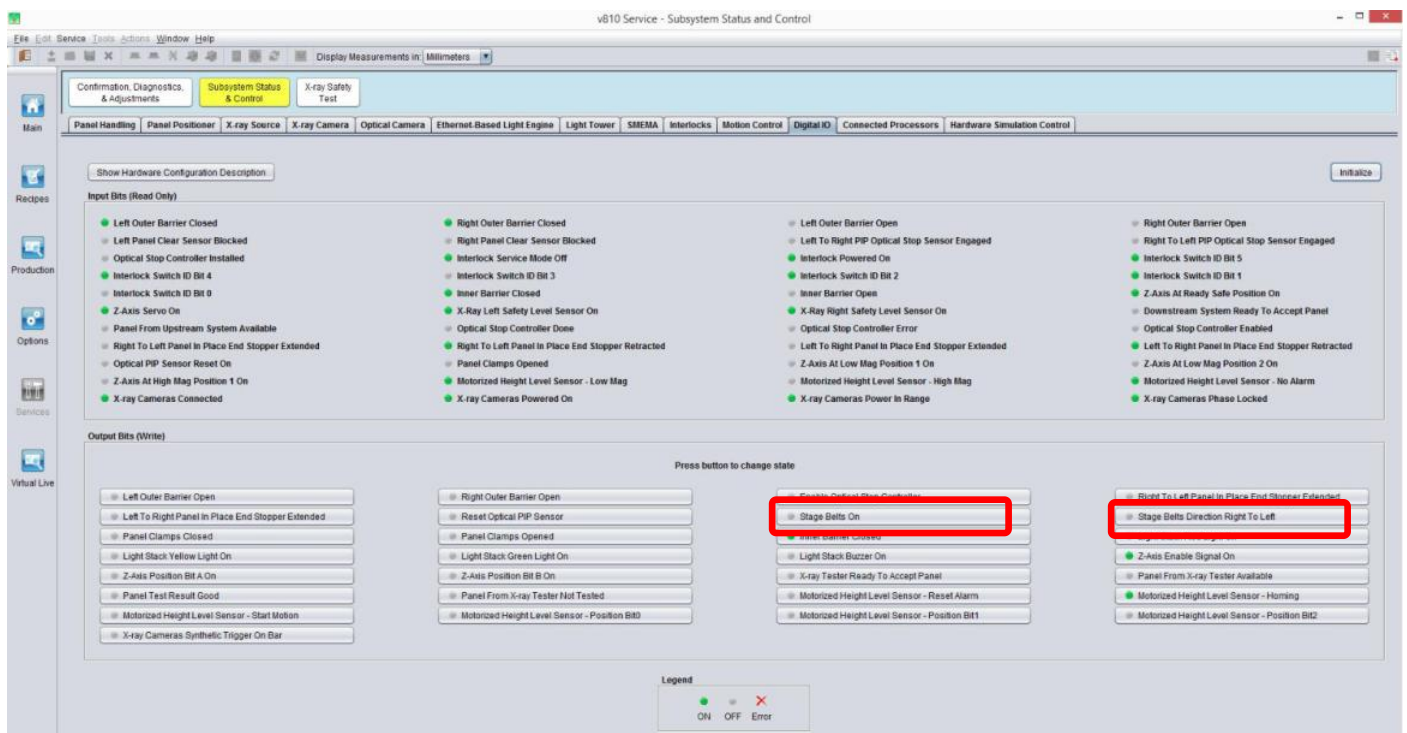


Figure 20 Digital IO (Belt Direction Switch)

17. Visual check Conveyor Belt Running Condition.
18. Repeat 14~16 if Conveyor Belt Running Condition fail
19. Toggle Off “Stage Belt On” at Digital IO tab
20. Toggle Off “ Stage Belt Direction Right to Left” at Digital IO tab

PIP Sensor Flag Verification and Adjustment

21. At Digital IO, Extend Both PIP & Toggle Belt Motion ON

a) Visual Check PIP Flag contain any friction between Conveyor Belt while Belt is running

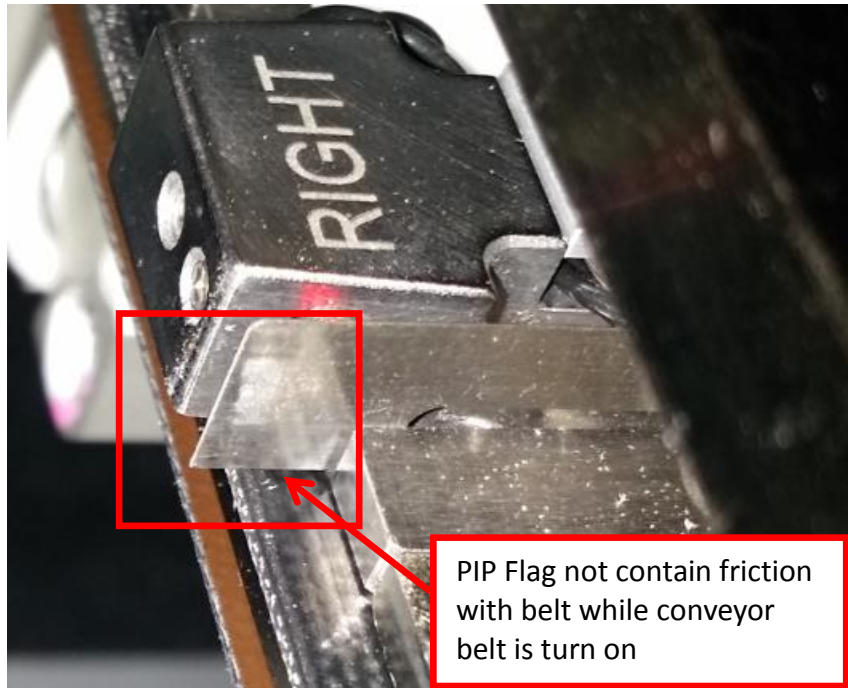


Figure 21

b) Check Digital IO Input Bit Status by refer to below Table

PIP Sensor	Left PIP	Right PIP
Status	No Engage	No Engage

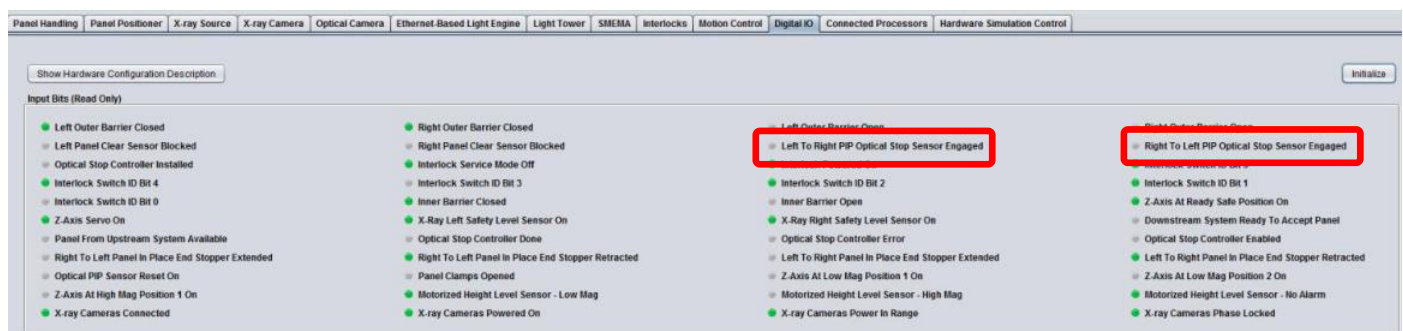


Figure 22 Digital IO

22. Remain at Digital IO Tab,Toggle “Stage Belts Direction Right to Left “ while “Stage Belts On” remain ON

23. Repeat Step 20 for another verification

ViTrox	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Fix Rail AWA Conveyor Belt Removal and Replacement Procedure	Effective Date	19 April 2016

24. Perform below Step if Step 20 Verification fail,else direct to Step 25

- a) Loosen 2 x Screws & remove Mount Flag as shown in Figure 23

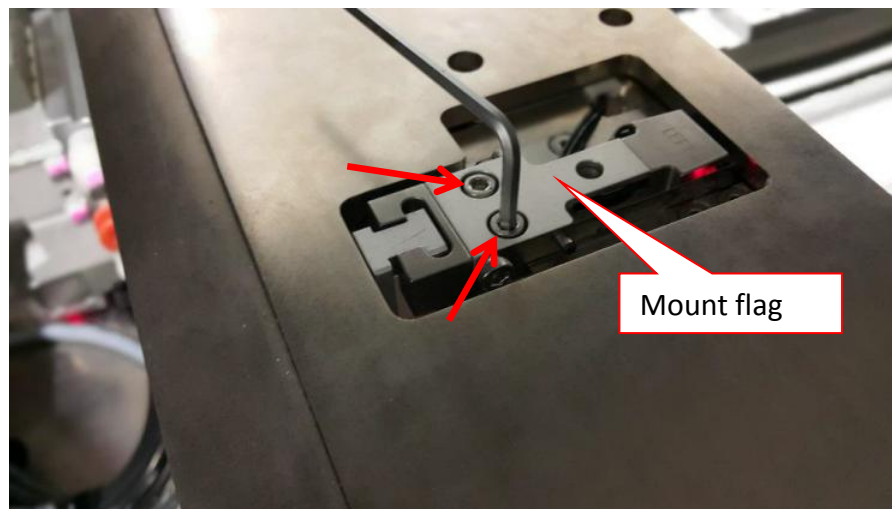


Figure 23 Mount Flag Removal

- b) Loosen Mount Flag's screw Gently align PIP Flag upwards

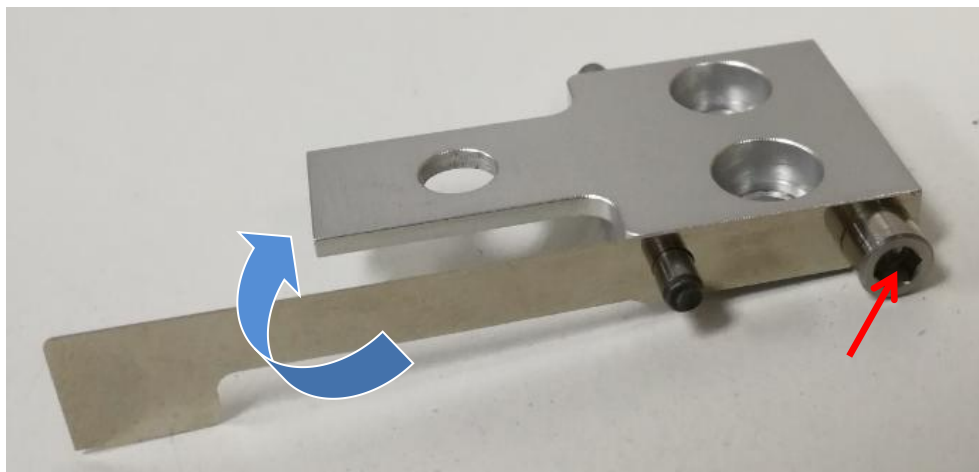


Figure 24 PIP Flag Alignment

- c) Assemble the Mount Flag and tighten 2 x Screws as Figure 23

25. Toggle Off "Stage Belt On" at Digital IO tab

26. Toggle Off " Stage Belt Direction Right to Left" at Digital IO tab

27. Remove all tools

28. Close all access door

29. Start machine as usual